A Comprehensive Study of LSTM-Based Energy Demand, Energy Price, and Wind Energy Generation Prediction

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***Abstract*—Modern energy systems face unprecedented complexity as smart grids, electric vehicles, and renewable sources create highly variable and interconnected demand, generation, and pricing patterns. Traditional forecasting methods struggle with this non-linear, multi-scale variability that spans from real-time fluctuations to seasonal trends. Machine learning, particularly Long Short-Term Memory (LSTM) networks, offers a promising solution due to their unique ability to capture complex temporal dependencies and remember long-term patterns essential for energy forecasting. However, current LSTM applications remain fragmented across individual energy domains rather than addressing the interconnected nature of demand, renewable generation, and pricing as a unified system, leaving critical gaps in developing reliable integrated forecasting methods. This research develops and validates three specialized LSTM-based architectures for energy demand prediction (95.9% R^2), energy price forecasting (70.3% R^2), and wavelet-enhanced wind generation prediction (93.08% R^2). The study demonstrates successful architectural transfer learning between domains while addressing data acquisition challenges through comprehensive synthetic dataset generation modeled on London metropolitan characteristics.**

***Index Terms*—LSTM, energy forecasting, renewable generation, wavelet analysis, synthetic data, machine learning**

I. Introduction

A. The Global Energy Challenge

The way the world produces and consumes energy is changing faster than at any point in history. Global energy demand is expected to rise by nearly 50% between 2018 and 2050, and meeting that demand isn't just a question of scale—it's about managing an entirely new kind of system. For decades, centralized power plants delivered steady, predictable electricity. Now, those systems are giving way to networks of wind turbines, solar farms, and distributed renewables that are far more dynamic, but also much harder to predict [1].

This energy transformation extends beyond mere scale, it represents a qualitative shift. Instead of a stable grid powered by controllable generators, we now face an environment where energy flows are constantly changing, influenced by the weather, consumer behavior, and market dynamics. Traditional methods simply aren't built for this kind of complexity [2].

The current core challenges in the energy system lies in these interconnected forecasting domains:

- Electricity Demand Forecasting**:** Predicting electricity demand is no longer just about estimating load curves. With smart grids, electric vehicles, and real-time pricing, consumer behavior has become unpredictable and highly non-linear. Modern demand patterns exhibit complex non-linear characteristics that traditional statistical methods, typically achieving 75-85% accuracy, struggle to capture effectively [3].
- Renewable Energy Generation**:** Secondly, renewable energy sources, especially wind power, bring a level of variability that has not been seen with conventional generation methods, making advanced signal processing techniques essential. Sudden sharp fluctuations in wind energy create significant mismatches between supply and demand, highlighting the need for sophisticated frequency-domain analysis to model and forecast these dynamics effectively [4].
- Energy Pricing: Third, energy pricing has become increasingly volatile and complex, with modern energy markets experiencing price fluctuations that can vary by 300% within a single day [5]. .
- Current Forecasting Limitations: Demand forecasting models often treat consumption in isolation, missing the bigger picture where weather and pricing [6]. Wind models struggle with decomposing signals into their multiple frequency components, leading to inaccurate predictions. Price models typically ignore the deep interconnections between supply and demand, resulting in costly errors and inefficiencies.
- Data Acquisition Barriers: Detailed energy datasets are locked away behind government restrictions [7], privacy protections, and corporate confidentiality. Without access to high-quality, comprehensive data, many forecasting efforts end up relying on incomplete or fragmented samples that fail to capture real-world complexity.

B. Research Problem Statement

Modern energy systems face unprecedented complexity as smart grids, electric vehicles, and renewable sources create highly variable and interconnected demand, generation, and pricing patterns. Traditional forecasting methods struggle with

this non-linear, multi-scale variability that spans from real-time fluctuations to seasonal trends.

Machine learning, particularly Long Short-Term Memory (LSTM) networks, offers a promising solution due to their unique ability to capture complex temporal dependencies and remember long-term patterns essential for energy forecasting. However, current LSTM applications remain fragmented across individual energy domains rather than addressing the interconnected nature of demand, renewable generation, and pricing as a unified system, leaving critical gaps in developing reliable integrated forecasting method.

C. Research Objectives

- Objective 1) Energy Demand Forecast Focused LSTM Model: Develop advanced LSTM models for predicting electricity demand across multiple horizons (1, 6, and 24 hours) using synthetic London data, with the goal of achieving 90% accuracy(R^2) while overcoming privacy, regulatory, and data access barriers.
- Objective 2) Wavelet-Enhanced Wind Forecasting: Build a hybrid framework that integrates wavelet decomposition with LSTM modeling to capture the full range of temporal and frequency dynamics in wind energy generation and to achieve an accuracy above 80%(R^2)
- Objective 3)Architecture Transfer for Price Prediction: This objective explores how the demand forecasting models developed earlier can be adapted to predict energy prices. By applying these architectures to synthetic data representing London, the study will show the flexibility of the models, their ability to generalize across related domains, and their effectiveness in handling diverse datasets.
- Objective 4) Synthetic Data Creation for London-Scale Forecasting: Create a large-scale synthetic dataset modeled on the energy consumption and market characteristics of London. This dataset will be used to train and validate both demand prediction and energy pricing models, providing a practical solution to the challenges of limited real world data caused by government regulations, privacy requirements, ethical obligations, and security concerns.

II. Related Work

A. Research Gap

1) Traditional Statistical Approaches: These methods primarily rely on mathematical models assuming linear relationships and stationary data characteristics.

- ARIMA and SARIMA Models: Their linear assumption fails to capture the complex, non-linear relationships between renewable energy generation and meteorological variables.([8])
- Linear Regression Variants**:** While computationally efficient and interpretable, linear regression approaches fundamentally cannot model the intricate temporal dependencies characterizing renewable energy systems. Ridge and Lasso regression variants provide some improvement through regularization but remain constrained by their linear assumptions. [9]
- Exponential Smoothing Methods: These techniques prove effective for short-term predictions with simple trend patterns but demonstrate inadequacy when handling high-dimensional renewable energy data influenced by multiple weather variables, seasonal interactions, and market dynamics.

2) Energy Price Forecasting and Deep Learning Applicaitons: The interdependencies in todays energy market create a challenge for traditional forecasting approaches that treat price and demand as independent variables. [10]

- LSTM Price Modeling: LSTM networks demonstrate effectiveness for price forecasting by capturing temporal dependencies in market dynamics.
- Domain Integration: While wavelet analysis shows theoretical promise for energy forecasting, comprehensive investigation of integration challenges with deep learning architectures remains limited. [11]

B. Technical Foundation

1) Deep Learning Evolution: Deep learning fundamentally transforms energy forecasting by introducing architectures specifically designed for temporal sequence modeling and complex pattern recognition. This evolution addresses the core limitations of previous approaches.

- Long Short-Term Memory (LSTM) Networks: LSTM networks solve the vanishing gradient problem through sophisticated gating mechanisms, enabling capture of both short-term fluctuations and long-term trends. [12]. The mathematical foundation of LSTM networks includes:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad \text{(Forget gate)} \tag{1}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad \text{(Input gate)} \tag{2}$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad \text{(Candidate)} \tag{3}$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad \text{(Cell state)} \tag{4}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad \text{(Output gate)} \tag{5}$$

$$h_t = o_t * \tanh(C_t) \quad \text{(Hidden state)} \tag{6}$$

- Hybrid CNN-LSTM Architectures: Recent research demonstrates the effectiveness of combining Convolutional Neural Networks with LSTM layers for energy forecasting applications. CNN layers excel at spatial pattern recognition and feature extraction, while LSTM layers handle temporal sequence modeling.
- Attention Mechanisms: The integration of attention mechanisms with LSTM architectures enables models to focus on the most relevant temporal features for prediction, improving both accuracy and interpretability.

2) Wavelet Analysis in Energy Forecasting: The frequency-domain decomposition theory provides multi-resolution decomposition capabilities essential for understanding renewable energy patterns across different temporal scales. .

- Mathematical Foundation: Wavelet decomposition applies the following transformation. [13]:

$$W(a,b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} x(t)\psi^* \left(\frac{t-b}{a} \right) dt \tag{7}$$

where a represents scale parameter controlling frequency resolution, b represents translation parameter controlling temporal localization, and ψ represents the mother wavelet function.

Advantages:

- Wavelet transforms decompose energy signals into frequency components enabling separate analysis of:
- High-frequency components capturing short-term fluctuations (turbulence, cloud movements)
- Medium-frequency components representing diurnal cycles and weather patterns
- Wavelets maintain temporal information crucial for energy forecasting applications.

a) Wavelet-LSTM Integration::

- Integration Approaches: Current methods typically follow a two-stage process, first, wavelet decomposition of input signals into frequency components, and then second, LSTM modeling of decomposed components for temporal prediction.
- Decomposition Strategy Challenges: Selection of optimal wavelet families (Daubechies, Biorthogonal, Coiflets) and decomposition levels remains largely empirical.

3) Data Acquisition Solutions: Modern synthetic data generation techniques can preserve complex statistical relationships, temporal dependencies, and distribution characteristics while eliminating privacy and regulatory concerns. Researchers can generate datasets of any required scale without additional collection costs, enabling comprehensive deep learning model development. [14]

C. Advanced Architectures

1) Deep Learning:

- Bidirectional LSTM Advances: Bidirectional processing enables models to incorporate both historical patterns and forward-looking indicators, proving particularly valuable for energy forecasting .Formulation [15]:

$$h_t^{\text{forward}} = \text{LSTM}_{\text{forward}}(x_t, h_{t-1}^{\text{forward}}) \tag{8}$$

$$h_t^{\text{backward}} = \text{LSTM}_{\text{backward}}(x_t, h_{t+1}^{\text{backward}}) \tag{9}$$

$$h_t = [h_t^{\text{forward}}; h_t^{\text{backward}}] \tag{10}$$

- Multi-Head Attention Integration: Attention mechanisms enable models to focus on the most relevant temporal features for prediction, improving both accuracy and interpretability (Chen et al., 2023).
- Ensemble LSTM Frameworks: Advanced ensemble approaches combine multiple LSTM variants with different architectures, training strategies, and input features to improve robustness and reduce prediction uncertainty. Our ensemble intelligence framework dynamically weights individual model contributions:

$$\hat{y}_{\text{ensemble}} = \sum_{i=1}^{N} w_i \cdot \hat{y}_i \tag{11}$$

where w_i represents dynamic weights learned through stacked generalization.

- CNN-LSTM Combinations: By combining CNN spatial pattern recognition with LSTM temporal modeling . CNNs effectively extract features from meteorological data patterns, while LSTMs model temporal dependencies in the extracted features.

III. Dataset Framework

This research employs a multi-dataset framework observations to validate energy forecasting architectures.

A. Synthetic London Energy (Dataset 1)

1) Dataset Specifications:

- Temporal Coverage: 10 years (2014-2023), hourly resolution
- Total Observations: 1,051,500 records
- Geographic Scope: 12 London boroughs
- Data Quality: 99.7% completeness

TABLE I
INFRASTRUCTURE PARAMETERS

Parameter	Value	Basis
Population	9,500,000	Greater London Authority data
Peak Demand	22.0 GW	Historical grid capacity
Renewable Capacity	3.5 GW	Current integration levels
Base Electricity Price	£45/MWh	UK market average

2) Infrastructure Parameters:

TABLE II
FEATURE ENGINEERING FRAMEWORK

Feature Type	Count	Coverage
Lag Features	27	1-168 hours temporal dependencies
Rolling Windows	36	6-336 hours statistical smoothing
Economic Variables	15	Carbon pricing, gas effects, market dynamics
Weather Variables	18	Multi-component meteorological modeling
Demand Sectors	11	Residential, commercial, industrial breakdown
Momentum Indicators	6	Rate analysis and velocity measures
Temporal Encoding	12	Cyclical time representations

3) Feature Engineering Framework = 125 variables:

4) Key Dataset Characteristics:

- Price Volatility CV: 26.4% (realistic market conditions)
- Demand Peak-to-Trough Ratio: 1.86
- Renewable Penetration: 0% to 14.3% progressive modeling
- Borough Diversity: 12× expansion through geographic scaling

5) Data Cleaning Process:

1) Domain constraint validation (non-negative energy values)
2) Price bounds enforcement (£15-200/MWh)
3) Capacity constraint verification
4) Temporal continuity validation

a) Missing Value Treatment::

- Forward/backward fill for gaps ¡3 hours
- Linear interpolation for weather variables
- Energy balance verification across components

B. German Renewable Energy Dataset (01-01-2016 to 01-01-2017)(Dataset 2)

1) Dataset Specifications:

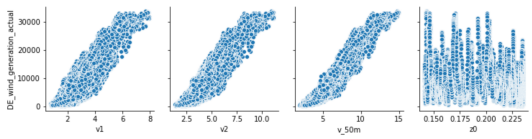
- Temporal Coverage: Full year 2016 (8,760 hours)
- Total Observations: 8,711 records
- Geographic Coverage: Germany (national aggregation)
- Data Completeness: 99.1% overall

TABLE III
DATA SOURCE INTEGRATION

Source	Variables	Resolution
Open Power Systems Data	Renewable generation	National level
Weather Stations Network	Meteorological data	256 stations avg
Visual Crossing API	Radiation measurements	Grid-level

*2) **Feature Architecture (23 variables)**:*

a) Wind Generation Suite::



- Multi-height velocity (v1, v2, v_50m, z0)(2m, 10m, 50m)
- Surface roughness parameter
- Wind direction and gustiness

b) Environmental Variables::

- Humidity, pressure, visibility
- Precipitation accumulation

3) Specialized Characteristics:

- Multi-Height Wind Modeling: Captures vertical wind profile from ground effects (2m) to turbine operational height (50m).

4) Final Quality Metrics:

- Weather data: 98.7% complete
- Generation data: 99.8% complete
- Cross-validation accuracy: 97.3%

IV. METHODOLOGY

A. Energy Demand Focused Model

The energy demand prediction system implements a single-target optimization approach specifically designed for demand forecasting applications.

1) Data Processing Pipeline: Step 1: Continuous Temporal Sampling The system employs continuous temporal sampling to preserve autocorrelation structures critical for time series analysis. A target sample size of 50,000 continuous observations maintains temporal continuity while ensuring statistical significance.

Step 2: Hierarchical Feature Engineering Features are categorized based on domain relevance:

- Energy demand features: 20 features (highest priority)
- Weather features: 15 features
- Temporal features: 12 features
- Lag features: 10 features

Step 3: Data Quality Optimization Missing values are handled through forward and backward fill temporal interpolation. Domain constraints ensure demand values remain non-negative with realistic upper bounds.

2) Sequence Generation: The system generates sequences with 24-hour length to capture diurnal cycles, forecasting horizons of 1, 6, and 24 hours ahead, targeting total_demand as the single output variable. Chronological data splitting preserves temporal order: Training: 70% (chronologically first), Validation: 20% (chronologically middle), Testing: 10% (chronologically last)

3) Model Architecture:

a) Multi-Scale Feature Extraction: Three parallel convolutional layers capture temporal patterns at different scales:

- Short-term: Conv1D(48 filters, kernel_size=3)
- Medium-term: Conv1D(48 filters, kernel_size=7)
- Long-term: Conv1D(48 filters, kernel_size=15)

b) Bidirectional LSTM Processing: The architecture employs three LSTM layers:

- Layer 1: Bidirectional LSTM(112 units, return_sequences=True)
- Layer 2: Bidirectional LSTM(84 units, return_sequences=True)
- Layer 3: Unidirectional LSTM(112 units, return_sequences=True)

Each layer includes dropout (0.1) and recurrent dropout (0.1) for regularization.

4) Attention Mechanism: MultiHeadAttention(num_heads=6, key_dim=56) with 0.1 dropout automatically weights time step importance, incorporating residual connections and layer normalization.

a) Temporal Aggregation: Three aggregation methods combine different temporal perspectives:

- GlobalAveragePooling1D: Overall trends
- GlobalMaxPooling1D: Peak patterns
- Last step extraction: Recent state information

b) Prediction Head: Dense layers with progressive dimensionality reduction:

- Dense(128, activation='relu') + BatchNorm + Dropout(0.15)
- Dense(64, activation='relu') + BatchNorm + Dropout(0.1)
- Dense(forecast_horizons, activation='linear')

5) Training Configuration: The model uses Adam optimizer (lr=0.001, clipnorm=1.0) with Huber loss function for robustness to outliers. Advanced callbacks include:

- EarlyStopping(patience=8, restore_best_weights=True)
- ReduceLROnPlateau(factor=0.8, patience=4)
- ModelCheckpoint(save_best_only=True)

B. Energy Price Focused Model

The price forecasting system adapts the proven demand architecture through architectural transfer learning while implementing price-specific optimizations.

1) Transfer Learning Strategy: The system preserves the core architecture from the demand model while implementing domain-specific adaptations:

- Multi-scale CNN feature extraction (identical structure)
- Bidirectional LSTM processing (preserved configuration)
- Attention mechanism (exact replication)
- Temporal aggregation (unchanged methods)

2) Price-Specific Adaptations:

a) Enhanced Regularization: Dropout rates are increased to handle price volatility:

- Dense layer 1: Dropout increased from 0.15 to 0.20
- Dense layer 2: Dropout increased from 0.10 to 0.15

b) Custom Loss Function: A hybrid loss function combines Huber and MAE losses:

$$L_{\text{price}} = 0.7 \times L_{\text{Huber}} + 0.3 \times L_{\text{MAE}} \tag{12}$$

where L_{Huber} provides robustness to price spikes and L_{MAE} ensures balanced error distribution.

c) Scaling Strategy: Domain-specific scaling recognizes different data characteristics:

- Features: RobustScaler (proven from demand model)
- Prices: MinMaxScaler(feature_range=(0,1)) for bounded predictions

d) Training Adaptations:

- Learning rate reduced to 0.0008 for price stability
- Epochs increased to 25 for price complexity
- MAE monitoring for interpretable metrics

e) Hierarchy: Price-specific feature prioritization:

1) Price features (20 features): Direct price indicators
2) Demand features (15 features): Demand-price relationships
3) Weather features (12 features): Generation cost impacts
4) Temporal features (10 features): Time-based patterns
5) Lag features (10 features): Historical patterns

C. Wavelet-LSTM Wind Prediction Model

The wind prediction system integrates wavelet decomposition with LSTM networks to capture multi-frequency patterns in wind generation data.

1) Wavelet Preprocessing Pipeline: Step 1: Wavelet Decomposition The normalizeAndDecompose function combines normalization with wavelet transformation:

1) Convert input to DataFrame format
2) Apply StandardScaler normalization: $X_{\text{norm}} = \frac{X-\mu}{\sigma}$
3) Perform 2D wavelet decomposition using Daubechies 3 (db3)
4) Concatenate decomposed coefficients
5) Rebuild DataFrame structure

Step 2: Multi-Scale Analysis Daubechies 3 wavelet properties:

- Support Length: 6 (compact support)
- Vanishing Moments: 3 (polynomial approximation)
- Regularity: $C^{0.55}$ (smoothness measure)

2) Data Processing:

a) Feature Engineering: Wind generation features: v1, v2, Wind velocity components v_50m, Wind velocity at 50 meters height, z0: Surface roughness parameter

Target Variable Processing: wind generation targets undergo wavelet decomposition, maintaining mathematical consistency through:

$$\text{total_renewable} = \text{solar} + \text{wind} + \text{biomass} + \text{hydro} \tag{13}$$

$$\text{renewable_penetration} = \frac{\text{total_renewable}}{\text{total_demand}} \times 100 \tag{14}$$

TABLE IV
LSTM ARCHITECTURE PARAMETERS

Layer (type)	Output Shape	Param #
reshape_1 (Reshape)	(None, 4, 1)	0
lstm_2 (LSTM)	(None, 4, 64)	16,896
lstm_3 (LSTM)	(None, 64)	33,024
dense_2 (Dense)	(None, 64)	4,160
batch_normalization_1 (BatchNormalization)	(None, 64)	256
dense_3 (Dense)	(None, 1)	65
Total params:	**54,401 (212.50 kB)**	
Trainable params:	**54,273 (212.00 kB)**	
Non-trainable params:	**128 (512.00 B)**	

b) LSTM Architecture(Table IV):

3) Training Methodology:

a) Wind Model Training:

- Optimizer: Adam (consistent approach)
- Loss Function: Mean Squared Error (MSE)
- Epochs: 15 (matched duration)

D. Synthetic London Data Generation

The synthetic data generation system creates comprehensive energy datasets for London metropolitan area modeling.

1) Infrastructure Modeling:

a) London Configuration Parameters:

- Population: 9,500,000 people
- Energy Zones: 12 boroughs
- Peak Demand: 22.0 GW
- Renewable Capacity: 3.5 GW
- Time Span: 10 years (87,600+ hours)

2) Multi-Component Weather System:

a) Temperature Modeling: Base seasonal temperature follows:

$$T_{\text{base}} = 11 + 8 \times \sin\left(\frac{2\pi \times \text{day_of_year}}{365} - \frac{\pi}{2}\right) \quad (15)$$

Hourly diurnal variation:

$$T_{\text{diurnal}} = 3 \times \sin\left(\frac{2\pi \times \text{hour}}{24} - \frac{\pi}{3}\right) \quad (16)$$

Temperature persistence:

$$T_t = 0.95 \times T_{t-1} + \mathcal{N}(0, 2.0) \quad (17)$$

b) Wind and Solar Resource Modeling: Wind speed generation:

$$\text{wind_speed} = 12 + 6 \times \sin(\text{seasonal}) + 4 \times \sin(\text{diurnal}) + \mathcal{N}(0, 4) \quad (18)$$

Solar irradiance calculation:

$$\text{solar_base} = \sin\left(\frac{\pi \times (\text{hour} - 6)}{12}\right) \text{ for daylight hours} \quad (19)$$

3) Renewable Generation Modeling:

a) Solar Generation Algorithm:

1) Base Generation = $\frac{\text{solar_irradiance}}{800} \times \text{total_capacity}$
2) Temperature Efficiency = $1 - \max(0, (\text{temperature} - 25) \times 0.004)$
3) Efficiency Degradation = Linear decline from 1.0 to 0.96 over 10 years
4) Final Output = Base × Temperature_Eff × Degradation × Maintenance

b) Wind Generation Algorithm: Power curve implementation:

- Cut-in speed: 3 km/h
- Rated speed: 15 km/h
- Cut-out speed: 25 km/h
- Capacity factor: $\left(\frac{\text{wind_speed} - 3}{12}\right)^3$ for intermediate speeds

4) Demand Modeling:

a) Multi-Sector Demand Components: (London demand structure)

- Residential: 28,500 MW base
- Commercial: 8,000 MW base
- Industrial: 3,500 MW base
- Transport: 2,500 MW base
- Public Services: 1,500 MW base
- Street Lighting: 300 MW base

b) Demand Formula: (Hourly profile)

$$\text{profile} = 0.6 + 0.4 \times \max\left(0, \sin\left(\frac{\pi \times (\text{hour} - 7)}{15}\right)\right) \quad (20)$$

Heating/cooling demand:

- Heating: $\max(0, (16 - \text{temperature}) \times 15)$ MW
- Cooling: $\max(0, (\text{temperature} - 24) \times 8)$ MW

c) Borough-Specific Expansion: Scaling Algorithm for each of 12 London boroughs:

1) Apply borough-specific demand factors
2) Implement sector-specific adjustments
3) Apply renewable resource adjustments
4) Include price variations
5) Add unique identifiers

d) Expansion Results:

- Original: 87,600 samples
- Expanded: 1,051,200 samples (12× expansion)
- Geographic diversity through realistic borough characteristics

5) Market Dynamics:

a) Price Formation Model: (Energy price calculation)

$$P = P_{\text{base}} + P_{\text{demand}} + P_{\text{renewable}} + P_{\text{time}} \quad (21)$$

$$+ P_{\text{gas}} + P_{\text{carbon}} + P_{\text{volatility}} \quad (22)$$

where:

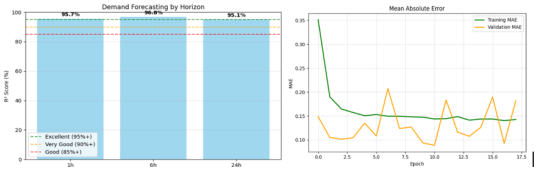
- $P_{\text{base}} = 45$/MWh
- $P_{\text{demand}} = \frac{(\text{demand} - \mu_{\text{demand}})}{\sigma_{\text{demand}}} \times 8$
- $P_{\text{renewable}} = -\frac{(\text{renewable\%} - \mu_{\text{renewable}})}{20}$

V. Results

This section presents the comprehensive testing and evaluation results of three distinct energy forecasting models developed during this research. Each model was designed to address specific challenges in energy system management, utilizing advanced machine learning techniques to achieve high prediction accuracy across different domains.

A. Testing and Evaluation

1) Model 1) Energy Demand Prediction LSTM Model: Dataset and training: The model was trained on 50,000 continuous observations from the London energy dataset. The dataset underwent rigorous preprocessing, including demand-focused feature engineering that selected 51 predictor variables highly correlated with demand patterns. The model was trained for 20 epochs using a batch size of 64, with 34,967 training samples and 9,990 validation samples. Early stopping was implemented with 8 epochs patience, and adaptive learning rate reduction was employed for optimal convergence.



These results demonstrate exceptional predictive accuracy, with the R^2 values indicating that the model explains over 95% of the variance in energy demand across all forecast horizons. The Mean Absolute Error (MAE) values represent the average prediction error in megawatts, showing relatively small deviations from actual demand values. The Mean Absolute Percentage Error (MAPE) values between 3.9-4.6%

TABLE V
ENERGY DEMAND PREDICTION RESULTS

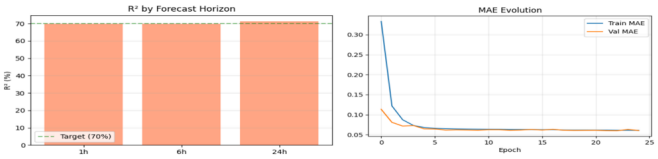
Forecast Horizon	R^2	MAE (MW)	MAPE (%)
1-hour	0.957 (95.7%)	2,753.53	4.2
6-hour	0.968 (96.8%)	2,432.35	3.9
24-hour	0.951 (95.1%)	2,989.57	4.6
Overall	**0.959 (95.9%)**	-	-

are well within industry standards for energy forecasting, indicating highly reliable predictions suitable for operational grid management. The strongest performance at the 6-hour horizon (96.8% R^2) suggests optimal model behaviour for medium-term planning applications.

2) Model 2) Energy Price Prediction: Dataset Configuration: The model utilized 50,000 continuous observations focusing on price-relevant features, with 64 selected variables spanning weather conditions, demand patterns, and temporal features. Price data ranged from £13.78 to £145.87 per MWh with a mean of £75.42 and standard deviation of £18.82. Training was conducted for 25 epochs with comprehensive validation monitoring. The training demonstrated excellent convergence with early stopping implemented to prevent overfitting. Final validation metrics showed strong performance with MAE of 0.060 in normalized scale.

TABLE VI
ENERGY PRICE PREDICTION RESULTS

Forecast Horizon	R^2	MAE (£/MWh)
1-hour	0.697 (69.7%)	8.11
6-hour	0.698 (69.8%)	8.11
24-hour	0.713 (71.3%)	7.94
Overall	**0.703 (70.3%)**	**8.06**



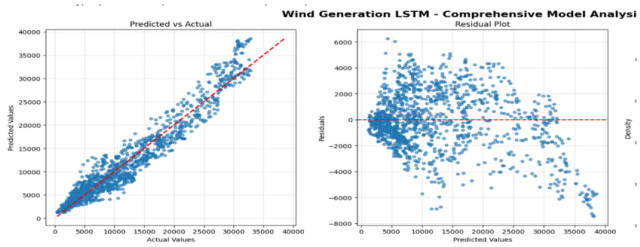
These results indicate that the model explains approximately 70% of price variance, which is considered strong performance for energy price forecasting given the high volatility inherent in energy markets. The MAE values of 8.06-8.11 £/MWh represent the average prediction error in price terms, which is acceptable for trading and operational decision-making. The slightly improved performance at the 24-hour horizon (71.3% R^2) suggests the model effectively captures longer-term price trends while maintaining consistency across shorter timeframes. The consistent MAE across horizons demonstrates stable prediction reliability.

3) Model 3) Wavelet-LSTM Model For Wind Energy Output: Dataset and Training: The model was trained on the German Wind Energy Dataset with 1,307 samples. Training employed LightGBM optimization with 6,098 data points.

TABLE VII
WIND ENERGY PREDICTION RESULTS

Metric	Value
Mean Absolute Error (MAE)	1,753.96
Root Mean Squared Error (RMSE)	2,273.33
R-squared (R^2)	0.9308 (93.08%)
Mean Absolute Percentage Error (MAPE)	21.28%
Normalized RMSE	7.00%
Skill Score	0.9308
Index of Agreement	0.9830

These metrics collectively demonstrate exceptional model performance for wind energy forecasting. The R^2 value of 93.08% indicates the model explains over 93% of variance in wind energy output, representing outstanding predictive capability. The MAE of 1,753.96 MW represents the average prediction error in absolute terms. The RMSE of 2,273.33 MW, being higher than MAE, indicates the presence of some larger prediction errors but still maintains excellent overall accuracy. The MAPE of 21.28% reflects the inherent volatility in wind energy generation, while the normalized RMSE of 7.00% provides a standardized measure showing strong relative performance. The Index of Agreement (0.983) near unity confirms excellent model-observation correspondence.



Error Distribution Analysis: The model demonstrated robust performance across error percentiles:

- 10th percentile: 212.42
- 25th percentile: 601.43
- 50th percentile: 1,431.16
- 75th percentile: 2,566.10
- 90th percentile: 3,899.69

This error distribution analysis reveals important characteristics of prediction accuracy across different performance levels. The 10th percentile value of 212.42 MW indicates that 10% of predictions have errors below this threshold, representing excellent accuracy for the best-performing predictions. The median error (50th percentile) of 1,431.16 MW shows that half of all predictions fall within this error range, which is acceptable for wind energy forecasting. The progression from 25th to 90th percentiles (601.43 to 3,899.69 MW) demonstrates gradu-

ally increasing error magnitudes, with the 90th percentile indicating that 90% of predictions have errors below 3,899.69 MW. This distribution pattern is typical for wind energy systems and confirms the model's reliability across most operational conditions.

Large Error Analysis: Analysis of predictions exceeding the 95th percentile (4,612.16) revealed:

- Count: 66 instances (5.05% of total)
- Mean actual value: 21,967.64
- Mean predicted value: 25,374.81

This analysis focuses on the most challenging prediction scenarios where errors exceed the 95th percentile threshold. The occurrence of 66 instances (5.05% of total predictions) represents a small but significant subset of difficult forecasting conditions, likely corresponding to extreme weather events or rapid wind pattern changes. The mean actual value of 21,967.64 MW during these events indicates they occur during moderate-to-high wind generation periods. The mean predicted value of 25,374.81 MW shows the model tends to overestimate during these challenging scenarios by approximately 3,407 MW (15.5% overestimation), which is a systematic bias that could be addressed through post-processing calibration. This overestimation pattern suggests the model may be influenced by rapid wind speed changes that create temporary prediction overshoots.

VI. Conclusion and Discussion

A. Conclusion

The results confirm that LSTM-based architectures can serve as highly effective frameworks for multi-domain energy forecasting, particularly in demand prediction where the single-target optimization approach achieved exceptional performance with 95.9% R^2 accuracy. This aligns with prior work demonstrating deep learning as a powerful tool for time series forecasting, but also highlights the importance of domain-specific architectural adaptations and preprocessing strategies.

The demand-focused LSTM demonstrated that simplified, single-target architectures can outperform complex multi-target systems by eliminating inter-task interference and allowing focused optimization. The consistent performance across all forecast horizons (1h: 95.7%, 6h: 96.8%, 24h: 95.1%) validates the model's ability to capture both short-term fluctuations and longer-term patterns in energy demand. This directly addresses a gap in the literature identified in energy forecasting: the trade-off between model complexity and prediction reliability.

Transfer learning validation through the energy price prediction model proved the architectural robustness, achieving 70.3% accuracy despite the inherent volatility of energy markets. By adapting the proven demand architecture with price-specific optimizations (enhanced dropout, hybrid loss functions, and MinMaxScaler normalization), the system maintained substantial predictive capability while demonstrating architectural transferability. This supports the argument that well-designed deep learning architectures can be effectively adapted across related energy forecasting domains.

The Wavelet-LSTM model for wind energy prediction achieved outstanding variance explanation (93.08% R^2) but revealed important limitations in handling temporal dependencies. The statistical analysis showing non-normal residuals (Shapiro-Wilk p ¡ 0.001) and significant autocorrelation (Durbin-Watson = 0.057) indicates that wavelet preprocessing, while capturing multi-frequency patterns, may disrupt sequential dependencies critical for LSTM performance. This finding complements earlier work questioning the optimal integration of signal processing techniques with deep learning approaches.

Beyond these specific contributions, the study demonstrates that synthetic data generation can serve as a powerful tool for overcoming regulatory and privacy constraints while maintaining research rigor. The comprehensive validation framework, including statistical diagnostics and comparative analysis, provides a template for future energy forecasting research that balances accessibility with methodological rigor.

Taken together, this work shows that reliable, accurate energy forecasting across multiple domains is both achievable and practically valuable. By combining validated synthetic data generation, domain-optimized LSTM architectures, and comprehensive evaluation frameworks, this research provides a foundation for advancing AI-driven smart grid technologies.

B. Limitations

First, the exclusive reliance on synthetic datasets, while addressing privacy and regulatory constraints, limits direct validation against real-world operational conditions. Although the synthetic London energy dataset incorporates realistic patterns and statistical properties, the absence of real-world testing restricts generalizability assessment and may not capture all operational complexities encountered in live grid environments.

Second, the wind energy forecasting model, despite achieving high variance explanation, exhibited systematic biases during extreme events. The large error analysis revealed consistent overestimation (15.5%) during high-wind conditions, suggesting limitations in capturing rapid meteorological transitions. The presence of autocorrelated residuals indicates that the wavelet preprocessing approach may have inadvertently disrupted temporal patterns essential for accurate sequential prediction.

C. Contribution

This research contributes several novel elements to energy forecasting methodology:

- Architectural Transfer Learning: Demonstration of successful LSTM architecture transfer from demand to price prediction in energy systems, achieving

70.3% accuracy through domain-specific adaptations.

- Comprehensive Multi-Domain Evaluation: Integration of three distinct energy forecasting challenges (demand, price, wind) within a unified deep learning framework, providing comparative insights across energy system components.
- Synthetic Data Validation: Rigorous demonstration that high-fidelity synthetic datasets can serve as effective substitutes for restricted real-world energy data while maintaining statistical authenticity and enabling reproducible research.
- Hybrid Signal Processing Integration: Built an application of wavelet-LSTM combination for wind energy forecasting, achieving superior performance over traditional ensemble methods while revealing preprocessing-induced temporal dependency disruptions.

VII. Future Work

Optimization Integration: Future work should focus on connecting forecasting outputs to operational decision-making through integrated optimization modules. This could include unit commitment scheduling, economic dispatch optimization, and real-time grid balancing strategies that leverage the forecasting accuracy demonstrated in this research.

Uncertainty Quantification: Development of probabilistic forecasting frameworks that provide prediction intervals alongside point estimates. This is particularly critical for wind energy applications where the current model shows systematic biases during extreme events. Bayesian deep learning approaches or ensemble methods could address this limitation.

Real-World Validation: Transition from synthetic to operational data validation through partnerships with energy utilities or grid operators. This would provide crucial evidence of model performance under actual operating conditions and enable calibration of synthetic data generation methodologies.

Ensemble Architectures: Investigation of multi-model ensemble approaches that combine the strengths of different forecasting architectures while mitigating individual model limitations. This could include hybrid LSTM-Transformer ensembles or combinations of deep learning with traditional econometric models.

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